2026-2027

CORNELL ROCKETRY TEAM



Sponsorship Prospectus

ABOUT US

We are ***Cornell Rocketry Team.***

Cornell Rocketry is a 55-member undergraduate **engineering project team** working with the College of Engineering at Cornell University to provide hands-on experience in aerospace engineering.

CRT designs, manufactures, and launches a fully **SRAD (student researched and designed)** high-powered hybrid rocket every year. Our competition is the **International Rocket Engineering Competition (IREC)**, the world's largest intercollegiate rocket competition. The primary goal is to launch our 20 ft tall rocket to a target apogee of 10,000 feet.

This year, we will travel to Midland, Texas to compete against over 160 other collegiate teams from 22 different countries.



1 student researched and designed **liquid rocket engine** - expected to fire in Spring 2027

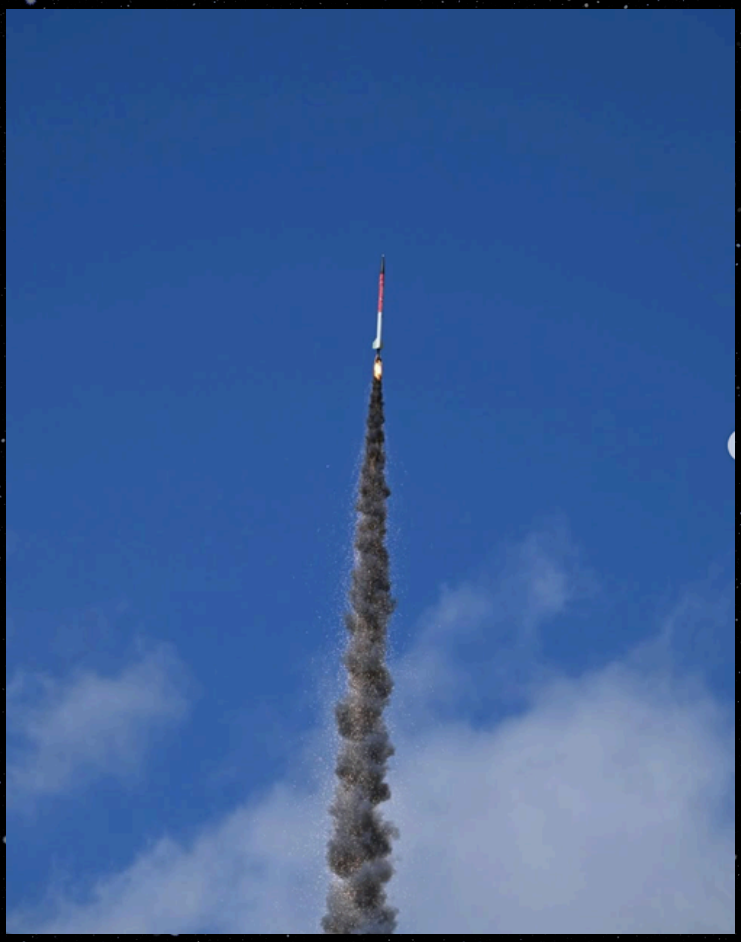
2 years of flying a fully student designed **hybrid** rocket

3rd place in our category at the **2025 and 2026 IREC**

6 **hotfires** of our SRAD propulsion system. 2 more expected this year

7 flights of our solid motor **testbed** rocket

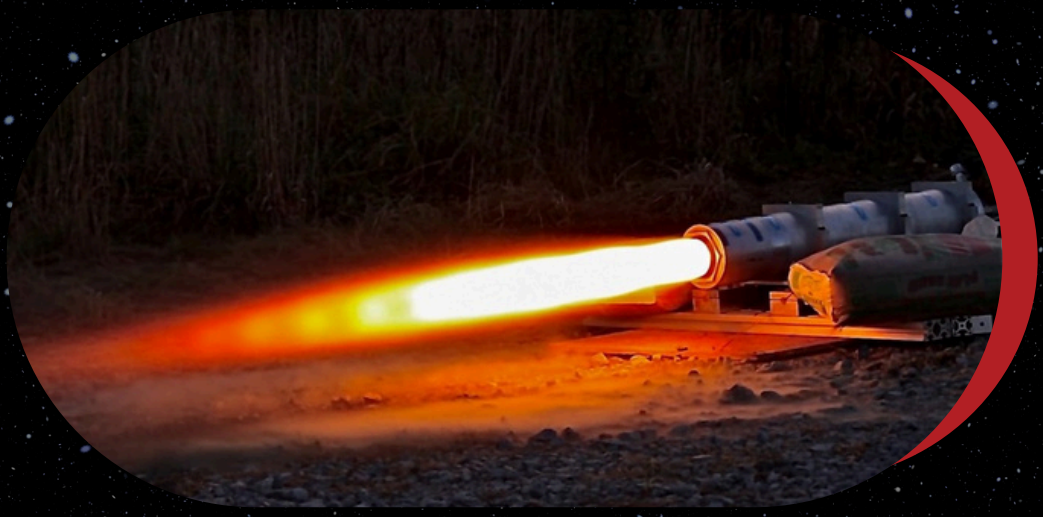
WHY CRT?



Our 6 subteams — Propulsion, Structures, Electrical, Software, Recovery & Payload, and Business — need your support to help us design and test rockets and to gain exposure to important workforce skills.

By supporting CRT, sponsors can gain visibility throughout Cornell and nationally, recruit promising, talented engineers from our team, and advertise through our social media channels. Donations will go towards rocket building materials, travel expenses to competitions, and safety equipment.

We also offer specific benefits depending on the amount donated - see our sponsorship tiers on the next page.



BENEFITS

The following are suggested tiers- contributions of any amount are appreciated.

	Bronze $500	Silver $1000	Gold $2500	Platinum $5000	Diamond $10000
General					
Newsletter	✓	✓	✓	✓	✓
Visibility at Cornell		✓	✓	✓	✓
National Visbility*			✓	✓	✓
Access to Designs				✓	✓
Tour of Workspace					✓
Info Session at Cornell					✓
Personal Thank You's					✓
Recruitment					
Access to Resume Book			✓	✓	✓
Access to Members**				✓	✓
Branding					
Logo on Website	✓	✓	✓	✓	✓
Logo on Rocket		✓	✓	✓	✓
Logo on Apparel		✓	✓	✓	✓
Public Thanks					✓

*Companies will gain widespread visibility through our national competitions

**Companies may reach out to CRT year-round for recruitment, new-product, or tech-talks.

HISTORY & FUTURE

Over the past 12 years, CRT has evolved from building commercial off the shelf rockets to entirely student build pressurized systems. The ultimate goal of the team is to continuously innovate and strive to achieve newer heights. That's why we're moving towards some of these projects this year:

- Tooling to create composite spin wound airframes in house
- 30 foot launch rail
- Liquid rocket engine - including TCA, injector, integration, tanks, and test stand
- FPGAs
- Computer vision projects
- Automated fill of oxidizer

...and more!



2024
1st 10k SRAD Solid
2nd Overall

2025
3rd 10k SRAD Hybrid

2026
3rd 10k SRAD Hybrid

This would not be possible without your support. We are excited to continue to innovate over the next 12 years!

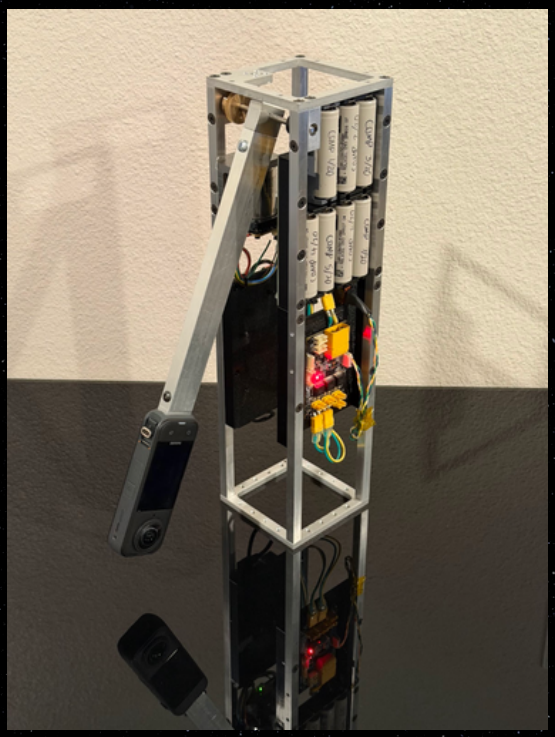
12

years as a student project team

SYSTEMS

The team designs and manufactures nearly every part of the rocket **in-house,** making use of a variety of lab spaces on campus. While competition is the major goal for the team, our primary focus is to provide all of our members with the opportunity to learn and grow as engineers with **hands-on experiences.** The systems highlighted here are not an exhaustive list of what students are working on.

Payload



Each year, the team flies a scientific **payload** onboard. The purpose changes from year to year, but our most recent payload was a **deployable 360 camera** to record parachute deployment in addition to ascent and descent of the rocket. A capture from the onboard camera's flight in June is shown below.

This year, students are continuing to develop this system while adding livestream capability.

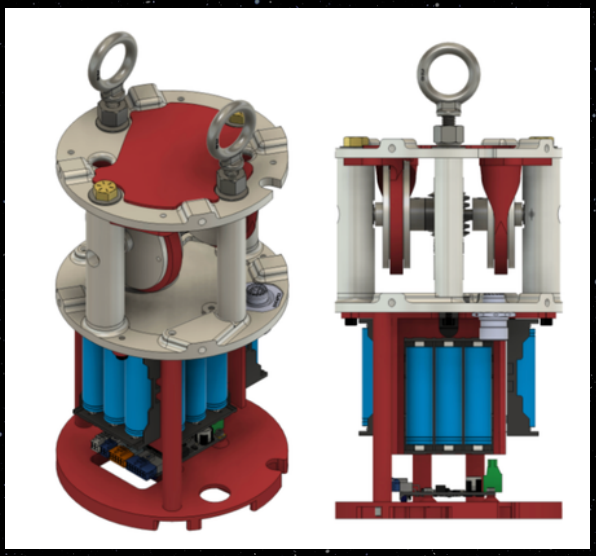


SYSTEMS

BLiMs

Students have been developing **BLiMs (Brake Line Manipulation System)** for the past 6 years, and are currently working on a comprehensive system redesign. BLiMs steers the rocket's main parachute, which is actually a parafoil, like a skydiver does - controlling pitch and yaw using cords attached to the parafoil.

The complex PID controller takes in accelerometer and GPS data to steer the rocket's descent to a previously set GPS waypoint.



Propulsion





CRT flies a **hybrid propulsion** system capable of **1250 lbf** and **20,000 Ns** of thrust. The entire N2O-HTPB hybrid propulsion system is Student Researched and Designed, including the fluids fill station, pressure vessel, bulkheads, quick disconnect, fuel grain, and test stand.

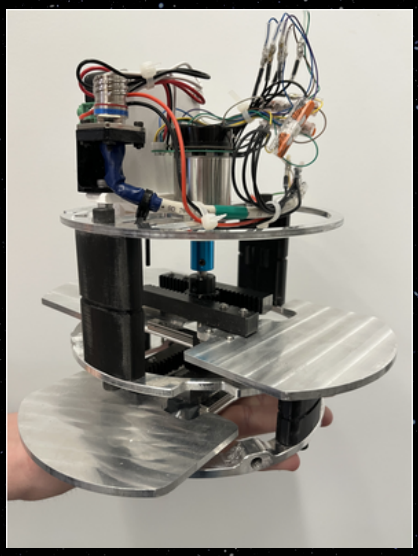
Electrical and Software additionally create **custom Fill Station PCBs** and software for use with pressurized systems.

SYSTEMS

Airbrakes

CRT began developing an **airbrakes** sytem in 2025. This system is intended to deploy leaves outside of the airframe of the rocket to increase drag coefficient and thus slow down the rocket's acceleration. This will help the team to hit the 10,000 ft apogee as closely as possible.

Airbrakes also has a student-developed controller/simulation. Both the software and the mechanism are being redesigned this year.



Avionics Bay



All of the **PCBs** designed for flight are designed and created in-house. These PCBs include a backplane, fluid sense board, flight computer, solenoid valve actuation board, and much more. The flight computer runs our flight software loop and commands every system onboard.

The enclosure is similarly redesigned every year to reduce torsion and ensure safe connection to the wiring harness.







THANK YOU



